

Country Analysis Meeting (CAM) 2025 - Analysis Guide

Jun 15, 2025

Table of contents

Welcome

Overview of the Countdown to 2030

The Countdown to 2030 for Women's, Children's and Adolescents' Health (The Countdown) initiative is a global collaboration involving academics from national, regional, and international institutions, UN agencies, the World Bank, and civil society organizations. The initiative tracks progress in Reproductive, Maternal, Newborn, Child, and Adolescent Health and Nutrition (RMNCAH+N), fostering advocacy and accountability through rigorous data analysis.

Key objectives include:

- Strengthening country-led data analysis and monitoring
- Fostering innovation and evidence generation through multi-country collaboration
- Enhancing global measurement and monitoring Improving policy and program communication

For more on Countdown to 2030 initiative, visit: [The Countdown website](#)

About the `cd2030.rmncah` R Package

The `cd2030.rmncah` R package and Shiny App were developed to support evidence generation and analysis of RMNCAH indicators.

Key features include:

- User-friendly interface for data management and analysis
- Tools for visualization and statistical summation
- Automated report generation

Acknowledgments

The Countdown acknowledges and appreciates the contribution of the following individual in developing the Stata codes that form the backbone of the `cd2030.rmncah` R package and Shiny App:

Part I

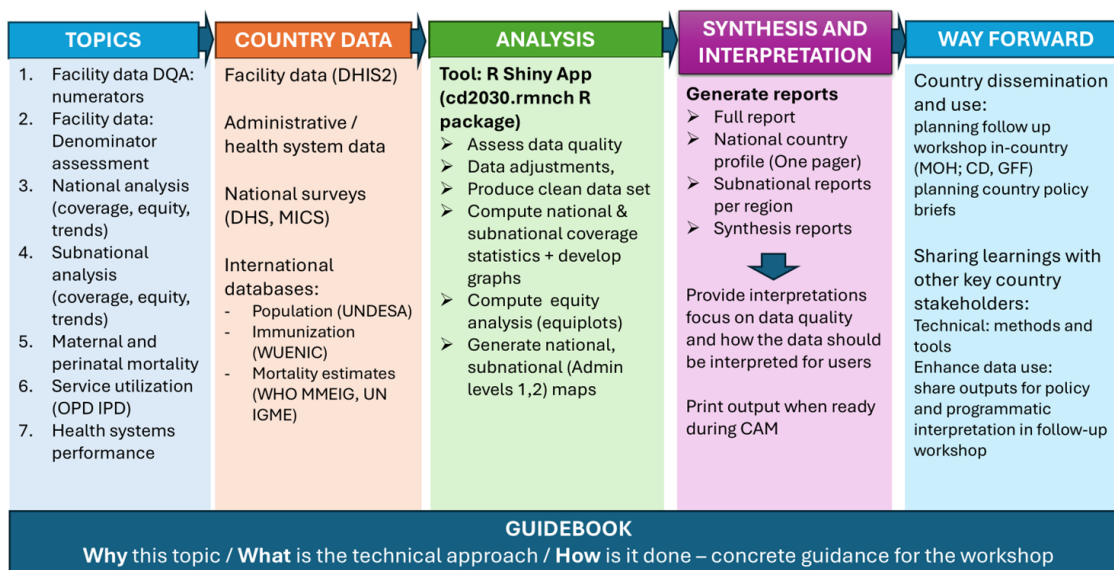
About this Guide

1 CD2030 CAM Approach

The CD2030 for Women's, Children's and Adolescents' Health, GFF, UNICEF, WHO, WAHO and other partners are collaborating to strengthen country-led progress and performance reviews, such as annual health sector reviews and midterm reviews of investment cases. This guidebook is for Countdown country analytical teams to develop a set of national and subnational estimates for key RMNCAH-N indicators, including equity, using five-year time series of routine data and survey results.

Much attention is paid to obtaining a clean data set with the necessary corrections and adjustments for known biases. Service coverage and equity, maternal and perinatal mortality, and health service utilization and systems performance are the main subjects, with a focus on monitoring national and sub-national targets, as well as global targets. The figure below shows the general overview of the CAM approach

COUNTDOWN CAM 2025 – GENERAL OVERVIEW OF THE APPROACH



1.1 Organization of the Guidebook

This guidebook is organized into seven sections, each focusing on a specific area of data analysis related to reproductive, maternal, newborn, child and adolescent health and nutrition (RMNCAH-N). The guidebook provides a comprehensive approach to analyzing routine health data and survey results, with an emphasis on data quality, coverage, equity, and health systems performance. The seven data analysis sections in this guidebook are:

1. Section 1: Health facility data quality assessment
2. Section 2: National Analyses (Coverage & Equity)
3. Section 3: Subnational analysis (Coverage & Inequality)
4. Section 4: Maternal mortality, stillbirths and neonatal mortality
5. Section 5: Curative health services utilization for sick children
6. Section 6: Health systems progress and performance
7. Section 7: Planning ahead for data use

These sections are designed to be used in a modular way, allowing countries to select the topics that are most relevant to their context and data availability.

Each section has:

- **Why/Rationale** - the scientific basis for the analysis;
- **Approach**- a step-by-step guide on how to conduct the analysis; and the
- **Implementation** - the use of the R Shiny App for data visualization and interpretation.

1.2 Data Sources

The Countdown CAM approach uses a variety of data sources, including:

- **Health facility data:** Routine health data collected from health facilities, including service coverage, health systems performance, and health service utilization.
- **Surveys:** Nationally representative surveys, such as the Demographic and Health Surveys (DHS) and Multiple Indicator Cluster Surveys (MICS), which provide data on health indicators, equity, and health service utilization.
- **Administrative data:** Data collected by government agencies, such as vital registration systems and health management information systems (HMIS), which provide information on health outcomes and service delivery.
- **Other data sources:** Other relevant data sources, such as census data, population estimates, and health financing data, which provide additional context for the analysis.

1.3 Expected outputs

- **Synthesis/poster reports** (.pdf, .doc files)
- **Full country report** (national/sub-national) (.pdf, .doc files) - to be downloaded in sections from the Shiny App and compiled into a full report with analysis outputs and interpretations.
- **Country analytical reproducible files** (.rds files)
- **Adjusted and or summarized data files** (.csv, .dta, xlsx files)

Country analytical reproducible files (.rds files) are the final output of the analysis, which can be used for further analysis and visualization. These files contain the cleaned and processed data, country specific analysis parameters as well as the results of the analysis, including coverage, equity, and health systems performance indicators.

2 Getting started

In the past, Countdown to 2030 (CD2030) analyses were conducted using Stata, relying on a series of .do files. These scripts have since been translated into an R package — **cd2030.rmncah** — and integrated into a user-friendly Shiny App. This application enables health data users to load data, conduct data quality assessments (DQA), perform analyses, and generate insightful outputs using pre-designed report templates.

The tool facilitates in-depth inspection of data quality and supports subnational analysis (up to the administrative level 2) for various indicators related to data quality and RMNCAH outcomes.

2.1 Software requirements

To use this innovative digital health analytics tool, the following are required:

- Install the latest versions of **R** and **RStudio**
- Install the **cd2030.rmncah** R package latest version-noted by the superscript suffix in the name of the App
Ensure access to all required (datasets) and (parameters)
- Install **RTools** (for Windows users, to support package compilation)

2.2 Installing R and RStudio

To begin working in R, it is necessary to install both R and RStudio. R is the underlying programming language, while RStudio provides a user-friendly interface that simplifies the development and execution of R scripts. Both are freely available and widely supported across platforms.

Note: R and RStudio are not the same. R is the programming language, while RStudio is an integrated development environment (IDE) that makes working with R easier.

To ensure smooth installation and functionality, it is recommended to install **R** first, followed by **RStudio**. This order is important as RStudio relies on R to function properly.

2.2.1 Step 1: Download and Install R

R is distributed via The [Comprehensive R Archive Network \(CRAN\)](#). Select your operating system from the homepage: Windows, Mac, or Linux.

Windows

1. Navigate to *Download R for Windows* and select the “base” option.
2. Click the first link (e.g., “Download R x.x.x for Windows”) to download the installer.
3. Run the installer and follow the prompts. *Administrator privileges may be required.*
4. R will be installed in your system’s Program Files, with a shortcut added to the Start menu.

Mac

1. Click *Download R for Mac* on the CRAN homepage.
2. Download the latest release package and run the installer.
3. The default installation settings are typically sufficient. You may be prompted to enter your system password.

Note

R is not a graphical software application but a programming environment. It is best used in conjunction with RStudio, which provides a consistent and user-friendly interface across operating systems.

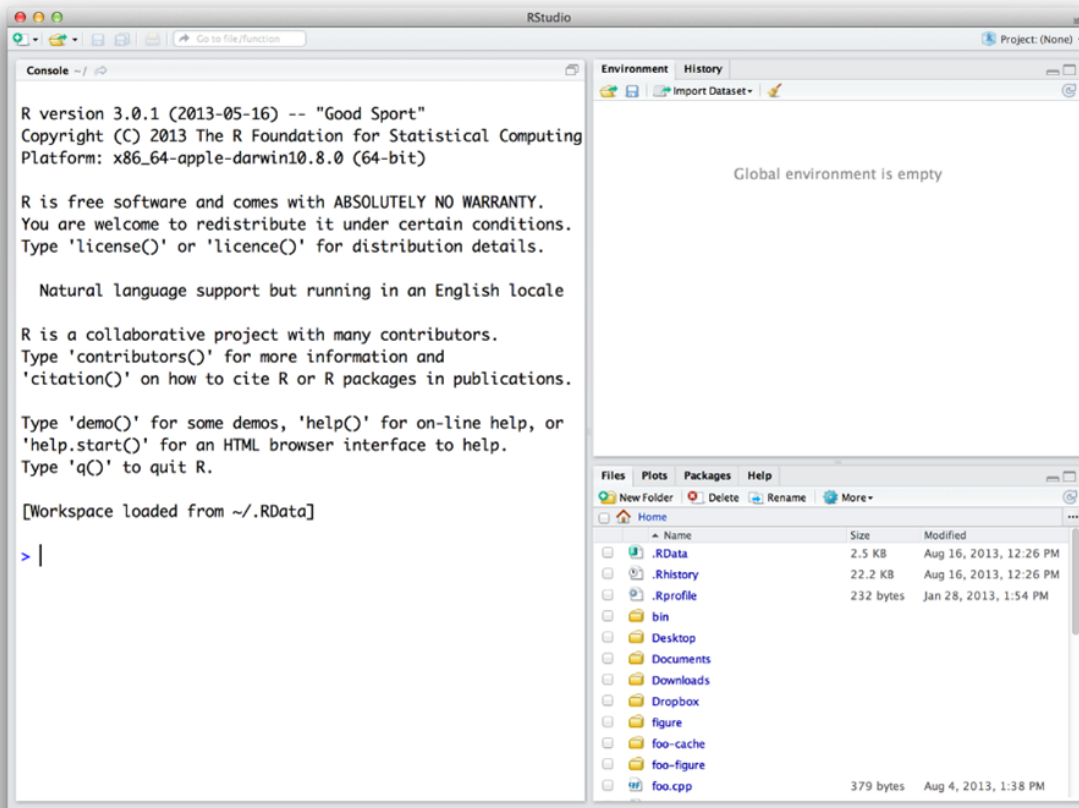
2.2.2 Step 2: Download and Install RStudio

RStudio is an integrated development environment (IDE) designed for R. It features a script editor, console, graphics viewer, and additional tools for package management, debugging, and file organization.

Download RStudio from [Posit Website](#)

Do I still need to download R? Even if you use RStudio, you’ll still need to download R to your computer. RStudio helps you use the version of R that lives on your computer, but it doesn’t come with a version of R on its own.

After installation, launch RStudio and begin interacting with R through its console and script windows.



2.3 Installing the CD2030 RMNCAH package

After installing R and RStudio, you may proceed with installing the Countdown2030 RMNCAH application, which is hosted on GitHub under the repository [cd2030.rmncah](https://github.com/cd2030/rmncah). The application is implemented as an R package and supports interactive dashboard generation via Shiny.

2.3.1 Installation via R Console

Stable Version

```
if (!requireNamespace("devtools", quietly = TRUE))
  install.packages("devtools")
devtools::install_github("aphrcwaro/cd2030.rmncah@v1.0.0")
```

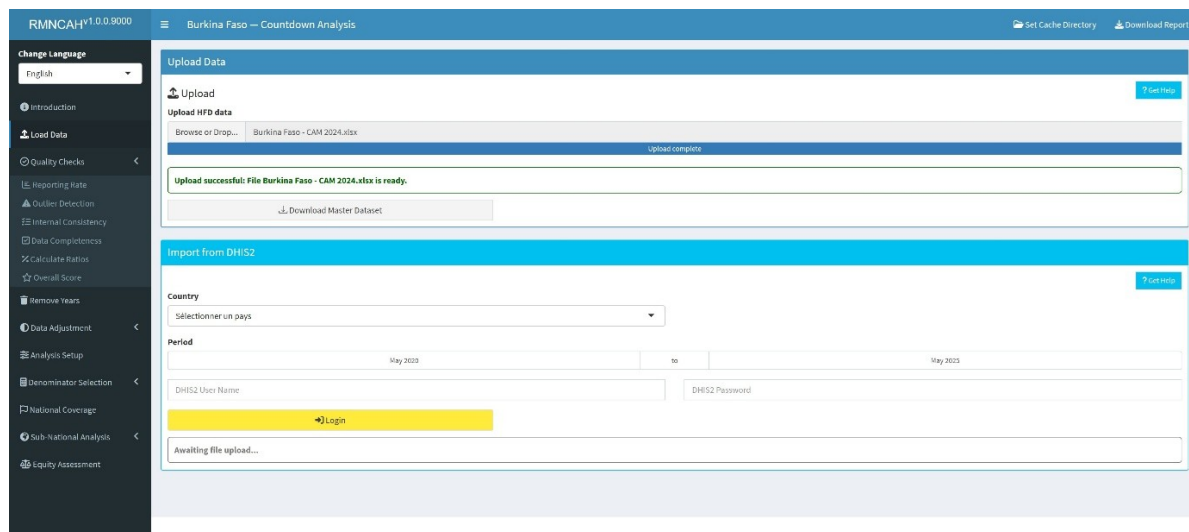
Development Version

```
if (!requireNamespace("devtools", quietly = TRUE))
  install.packages("devtools")
devtools::install_github("aphrcwaro/cd2030.rmncah")
```

2.3.2 Launching the Application

```
library(cd2030.rmncah)
dashboard()
```

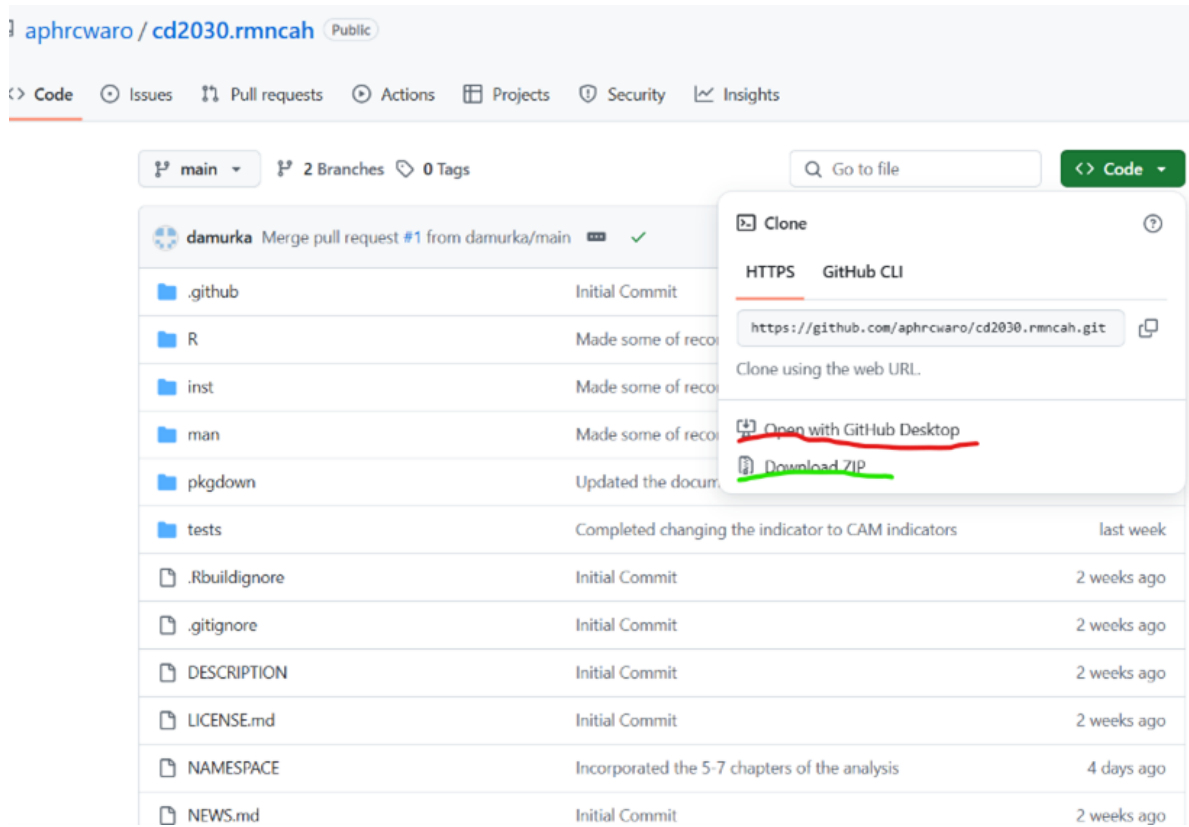
The Shiny dashboard will launch automatically in your default web browser.

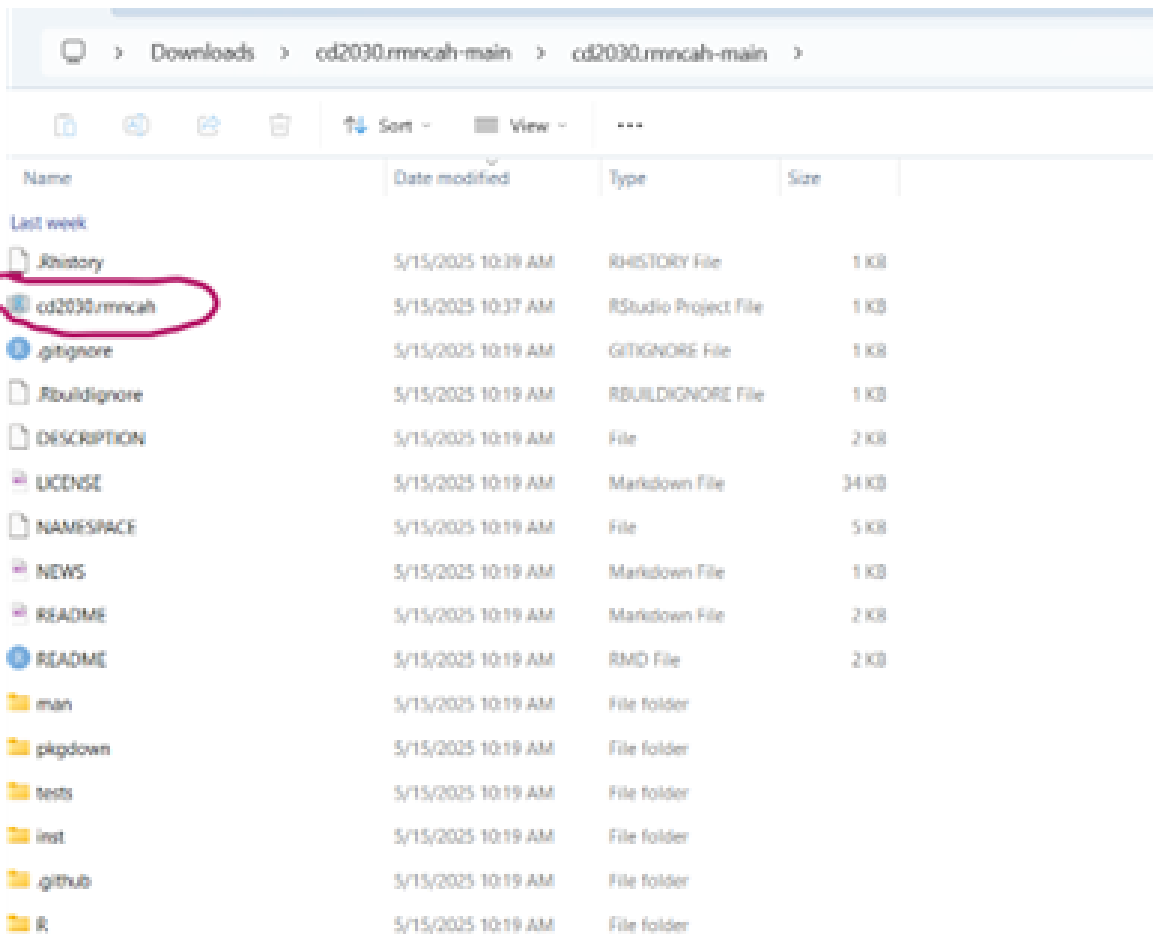


2.3.3 Alternative Installation via GitHub Desktop or Git or direct download

Advanced users with GitHub accounts may prefer to clone the repository directly. This method allows for:

- Version control
- Contribution to the codebase
- Inspection of the package structure
- Once cloned, open the .Rproj file in RStudio to set the working directory.





The screenshot shows a Windows File Explorer window with the address bar displaying the path: Downloads > cd2030.rmncah-main > cd2030.rmncah-main. The file list includes various files and folders, with the 'cd2030.rmncah' file highlighted by a red circle.

Name	Date modified	Type	Size
Last week			
.Rhistory	5/15/2025 10:19 AM	RHISTORY File	1 KB
cd2030.rmncah	5/15/2025 10:17 AM	RStudio Project File	1 KB
.gitignore	5/15/2025 10:19 AM	GITIGNORE File	1 KB
.Rbuildignore	5/15/2025 10:19 AM	RBUILDIGNORE File	1 KB
DESCRIPTION	5/15/2025 10:19 AM	File	2 KB
LICENSE	5/15/2025 10:19 AM	Markdown File	34 KB
NAMESPACE	5/15/2025 10:19 AM	File	5 KB
NEWS	5/15/2025 10:19 AM	Markdown File	1 KB
README	5/15/2025 10:19 AM	Markdown File	2 KB
README	5/15/2025 10:19 AM	RMD File	2 KB
man	5/15/2025 10:19 AM	File folder	
pkgdown	5/15/2025 10:19 AM	File folder	
tests	5/15/2025 10:19 AM	File folder	
inst	5/15/2025 10:19 AM	File folder	
github	5/15/2025 10:19 AM	File folder	
R	5/15/2025 10:19 AM	File folder	

To install and run:

```
library(cd2030.rmncah)
dashboard()
```

2.3.4 RTools for Windows

To compile packages from source, especially development versions, RTools must be installed. Ensure the Rtools selected corresponds and is compatible with your installed R version.:

- **R 4.2.0** → Rtools42
- **R 4.4.0** → Rtools44
- **R 4.5.0** → Rtools45

Rtools can be downloaded from [CRAN](https://cran.r-project.org/bin/windows/toolchain-links/)

2.4 Data requirements

2.4.1 Datasets required

To run the analysis efficiently, each country team will require to have a folder containing the following datasets:

1. Health facility data (.xlsx file)
2. UN Estimates
3. Survey data
4. UN mortality data
5. FPET data

These datasets will be provided to the country teams by the Countdown2030 team prior to the workshop.

2.4.2 Country specific analysis parameters

The following (parameters) will be required to run the analysis

3 App Features

3.1 Overview of App Features

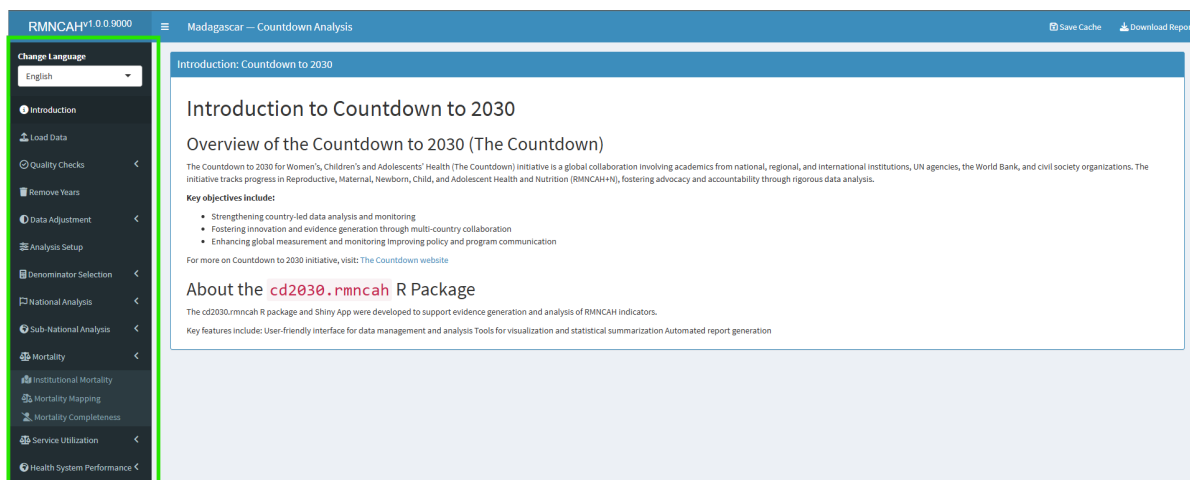
This section details the main components of the **cd2030.rmncah** Shiny App interface: the **Title Bar**, **Sidebar**, and **Main Panel (Body Content)**. Understanding these elements will help you navigate and use the app more effectively.

3.2 Title Bar

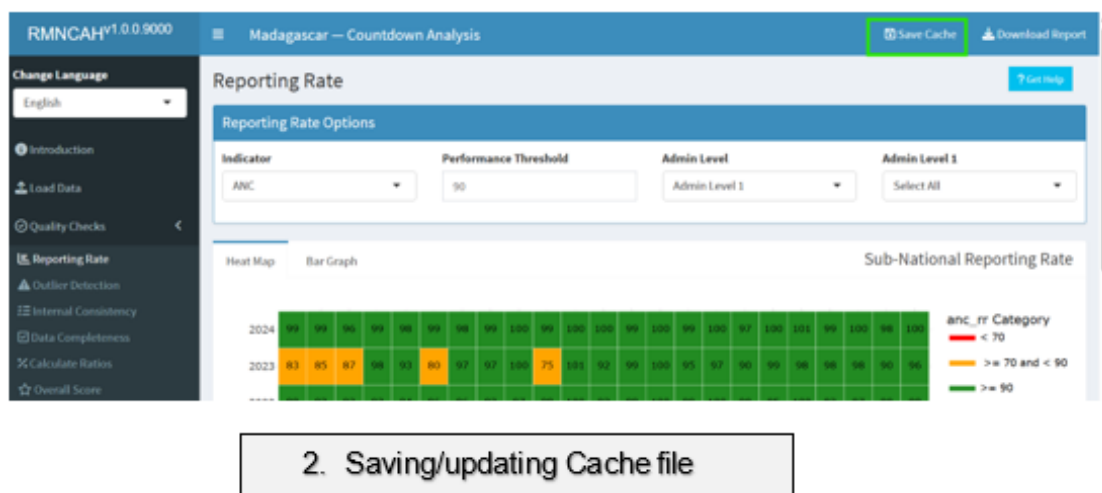
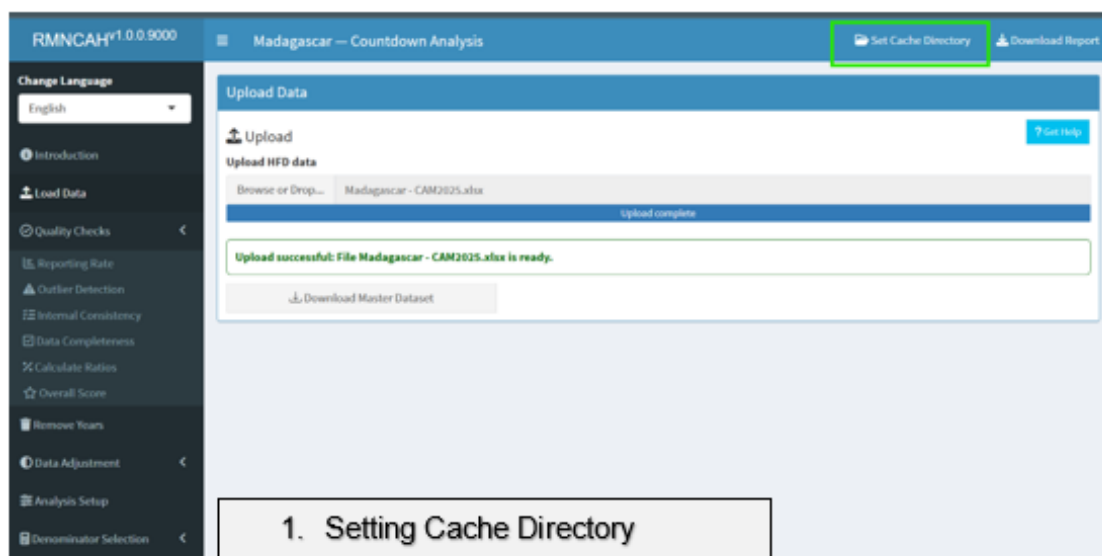
The Title Bar, located at the top of the app, provides key information controls. The elements of the Title Bar, such as “Set Cache Directory/Save Cache” and “Download Report,” are only displayed after data has been loaded.

3.2.1 Key Elements of the Title Bar

1. **Contextual Information:** Displays the current analysis context, such as the country of analysis (e.g., “Madagascar - Countdown Analysis”).



2. Cache Management

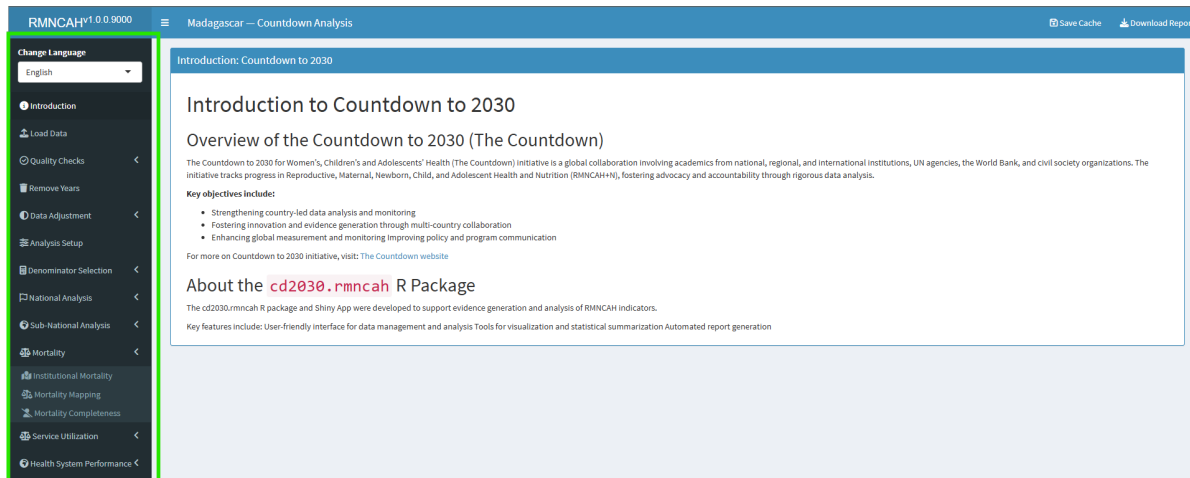


- **Set Cache Directory:** This button allows you to specify a directory for saving intermediate results and progress.
- **Save Cache:** This button allows you to save the progress of your analysis. This is useful for large analyses or for resuming a session later.
- NB:
 - The Cache button is enabled only once valid data has been uploaded into the App
 - Changing the file path of the Cache mid analysis would require you to set the cache again, losing any saved progress after the last save.

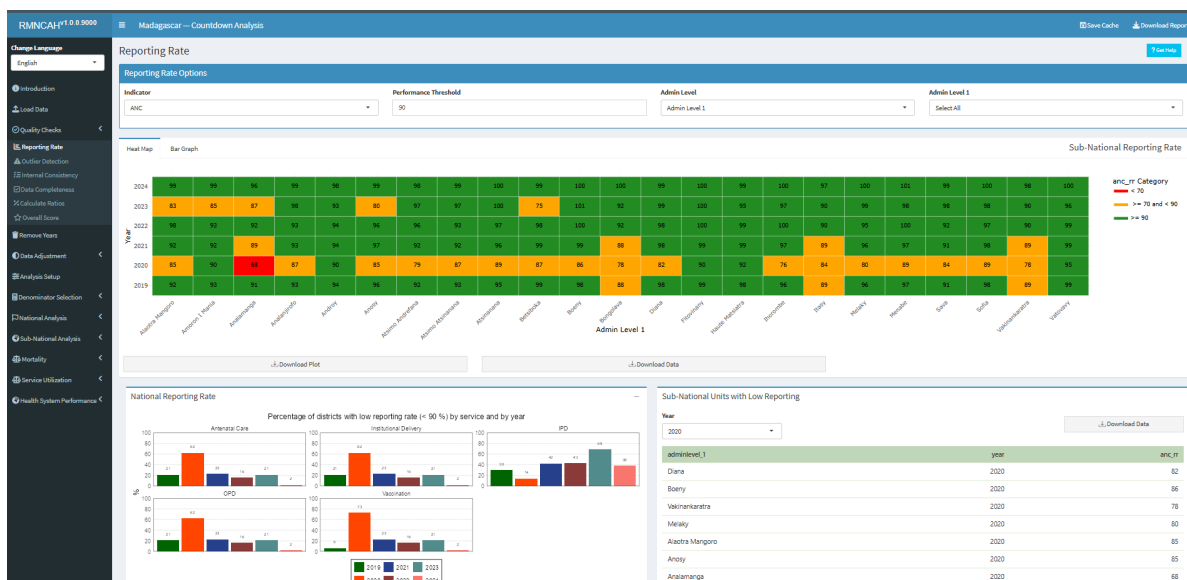
3.3 Sidebar

The Sidebar is located on the left side of the app and contains the main navigation elements. It provides navigation through the analysis workflow. The sections are arranged sequentially to guide you through the process.

3.3.1 Sidebar Navigation



1. **Introduction:** Provides a brief overview of the app's purpose and functionality.
2. **Load Data:** Allows you to upload and manage your datasets (e.g. Health Facility Datasets, Master datasets, and cache file).
3. **Data Quality:** Provides tools for assessing the quality of your data, including checks for missing values and inconsistencies. Subsections include:
 - Reporting Rate
 - Outlier Detection
 - Consistency Check
 - Data Completeness
 - Calculate Ratios
 - Overall Score



4. **Analysis Set-up:** Allows you to configure the analysis by selecting the data sources (UN Estimates, WUENIC estimates, and survey data). It also allows you to map the sub-national survey and mapping data to the uploaded data.

Analysis Setup

National Rates for Denominator Calculation (Based on the most recent survey)

ANC Survey (%) Pregnancy Loss (proportion) Twin Births (proportion) Neonatal Mortality (proportion)

Post-Neonatal Mortality (proportion) Stillbirth Rate (proportion) Prevalence Survey (%)

Survey Setup (percentage based on the most recent survey)

ANC Institutional Delivery Low Birth Weight Cesarean Section

Penta 3 Measles 1 BCG

Vaccines Survey Year Survey Data Start Year

Upload Survey Folder

Upload UN Estimates data Upload UN Mortality data Upload WUENIC estimates

Upload Survey Folder

Loading from cache

Subnational Manual Data Mapping (Optional)

Map Survey Data

Survey mapping has not been done

Map mapping has not been done

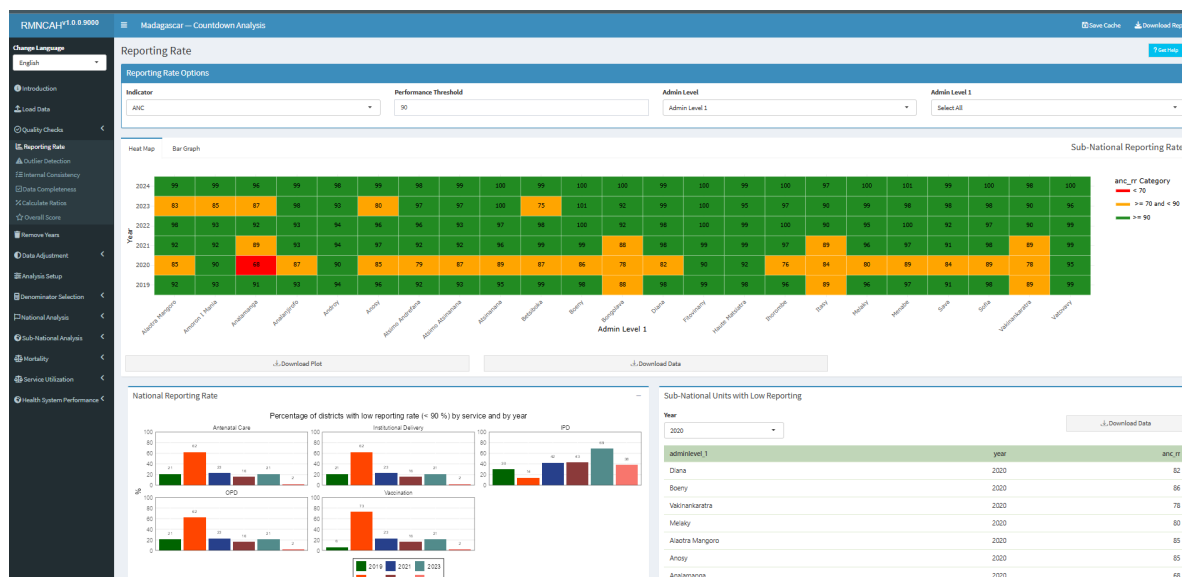
5. **National Analysis:** Provides tools for analyzing national-level data, including coverage and equity analysis.
6. **Sub-National Analysis:** For all sub-national analyses
7. **Mortality Analysis:** For all mortality analyses
8. **Service Utilization:** For Curative health services utilization for sick children analyses
9. **Health System Performance:** Health systems progress and performance analysis

3.4 Main Panel(Body Content)

The main panel is the central workspace where the analysis result, visualizations, and interactive elements are displayed.

3.4.1 Key Elements of the Main Panel

1. **Page Title:** Displays the current module or analysis step (e.g., “Reporting Rate”).



2. **Action Buttons:** Provides context-specific actions:

- **Download Report:** Helps you to download two reports: **Synthesis report** and the **Subnational one paper report**
- **Generate Report:** Downloads a report specific to the analysis section.
NB: This button is green in colour and only appears in the sections where the user can download the section reports.
- **Get Help:** Provides context-specific help and documentation for the current page.
- **Download Plot / Download Data:** Allows the user to download the displayed output and the data associated with the output, respectively.

4 Loading Health facility data

This section explains how to structure and upload data into the app, ensuring compatibility with the **Countdown Health facility data extraction template**

Please note that should you update your data during the workshop, ensure it is in the correct format before uploading to avoid errors.

4.1 Supported File Formats

The app supports uploading the following file types:

- **.xls, .xlsx** (Excel files)- The raw health facility dataset in the **Countdown data template**
- **.dta** (Stata files) - Master dataset downloaded from the app after validation/adjustment
- **.rds** (R Cached datafile) - The file containing the preloaded dataset, user adjustments and analysis parameters that has been > saved in the Cache directory. This will be the last saved file.

4.2 How to Upload Data

Step 1: Prepare your data file

Ensure your data is cleaned and structured according to the **Countdown Health facility data (HFD)** template by:

- Using the provided **HFD Standardized Template** to format your data correctly.
- Saving the file in a supported format: **.xls, .xlsx, .dta, or .rds**.

Step 2: Upload the File

1. Navigate to the **Upload Data** section of the app.

2. **Drag and drop** your file into the upload box, or click **Browse** > to select it manually in your directory.
3. For subsequent re-uploads (after the initial uploading of the **.xls**, **.xlsx**, **.dta** files and saving your progress in **.rds** file using the **Save Cache** button), do not re-upload the **.xls**, **.xlsx**, or **.dta** but the saved **.rds** file if you want to retain any changes made in your analytical files
4. The app will validate your file against the **Countdown template**.
 - If successful, a confirmation message will appear: ***“Upload successful: Your file is ready for analysis.”***
 - If errors are detected, an error message will indicate the issue.

4.3 Common Errors and How to Fix Them

Error Message	Cause	Solution
“Unsupported file format”	File type not supported	Save your file as .xls , .xlsx , .dta , or .rds .
“The following required columns are missing from the data: opv1”	Missing essential columns in the data	Add the missing column(s) to your dataset and ensure their values are valid.
“The following sheets are missing: Service_data_1, Service_data_2, Service_data_3, Reporting_completeness, Population_data, Admin_data”	Missing one or more required sheets in the file	Add the missing sheets to your file and ensure they conform to the template.
“Sheet Service_data_3 is empty”	The sheet exists but contains no data	Populate the sheet with valid data or remove the empty sheet.
“Key Columns”month” missing in Service_data_3”	A key column district, year, or month, is missing from the specified sheet	Add the missing column(s) to the sheet and ensure the data is structured correctly.
“Column name month must not be duplicated. Use .name_repair to specify repair.”	Duplicate column names in the dataset	Ensure all column names are unique. Rename or remove duplicate columns.

***Note:** If a key column (district, year, or month) is missing data in a row, that row will be excluded from the resulting dataset.*

4.4 Tips for a Successful Upload

- Always use the latest the **Countdown Health facility data template** > template to structure your data.
- Double-check column names, formats, and content before uploading.
- Save your file in a supported format and ensure it is UTF-8 encoded

Part II

Data Quality Assessment

5 Numerators Assessment

5.1 Rationale, Approach and implementation

Rationale: Scientific basis for the analysis

Routinely reported health facility data are an important data source for health indicators at facility and population level. The data are reported by health facilities on events such as immunizations given, or live births attended. As with any data, quality is an issue. Data needs to be checked to consider completeness of reporting by health facilities, identify extreme outliers and internal consistency. A standard reporting method for data quality allows assessment of progress over time.

Approach: Description of analytical steps

The analysis of the monthly data by district for 2019-2024 is used to assess annual data quality using the following standard indicators: *Reporting completeness, Outlier detection, Internal Consistency Data completeness, Ratios calculation, Overall quality score*

Implementation: Conducting analysis in the Shiny App

5.2 DQA Metrics calculation and interpretation

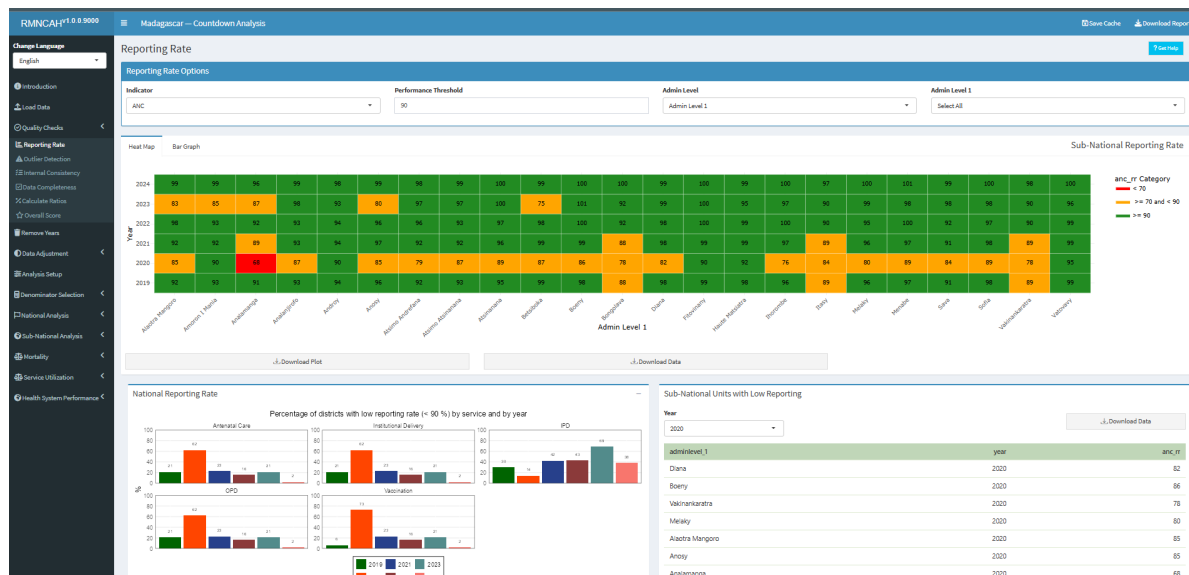
5.2.1 Reporting Completeness

Indicator	Numerator	Denominator
1a	N of monthly facility reports received	Total N of facility rep
1b	N of districts with at least 90% monthly reporting completeness in a year	Total N of districts
1c	N of districts with no missing values for any of the 4 forms in a year	Total N of districts

Statistics for 1a and 1b are based on the mean of 4 reporting forms (ANC, delivery, immunization, OPD)

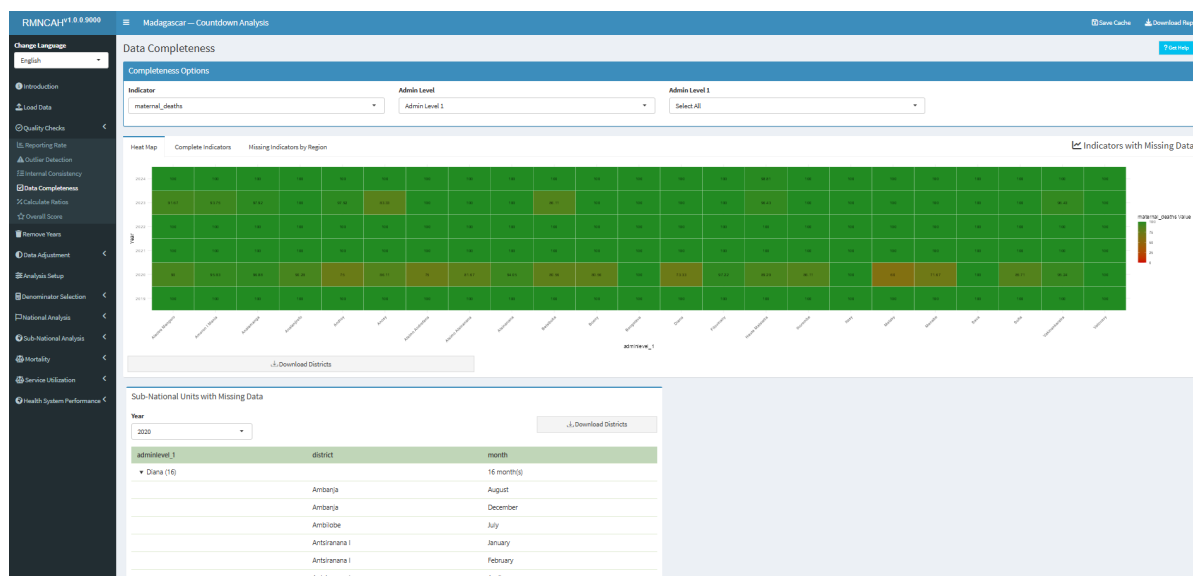
Within the Shiny App, you can select the indicator of interest, the reporting rate cutoff point (performance Threshold)- the default value has been set at 90% reporting rate.

You can also choose the level of your analysis (Admin Level (Regions) or District level). Below is an output for ANC reporting rate at the Admin Level 1.



The app will also produce a report with the results of the data quality assessment, including the numerators and denominators for each indicator, as well as the overall data quality score.

5.2.2 Data Completeness

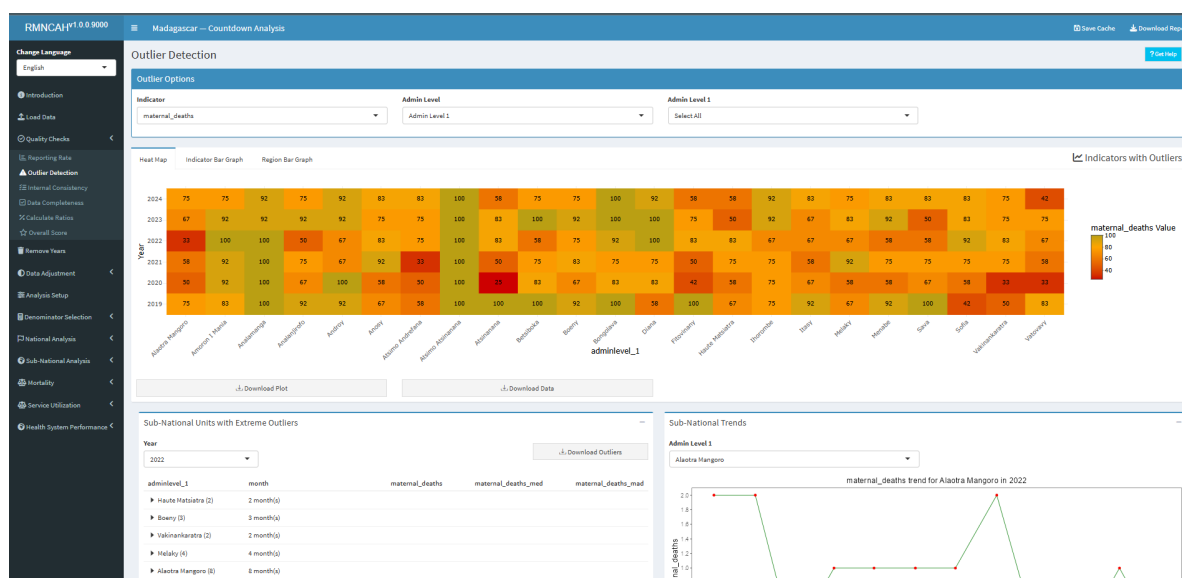


5.2.3 Outlier Detection

Indicator	Numerator	Denominator
2a	N of monthly values that are not extreme outliers in a specific year	Total N of monthly values (N of districts x 12 months)
2b	N of districts with no extreme outliers in a specific year	Total N of districts

Outliers are identified statistically; definitions may vary depending on methods used.

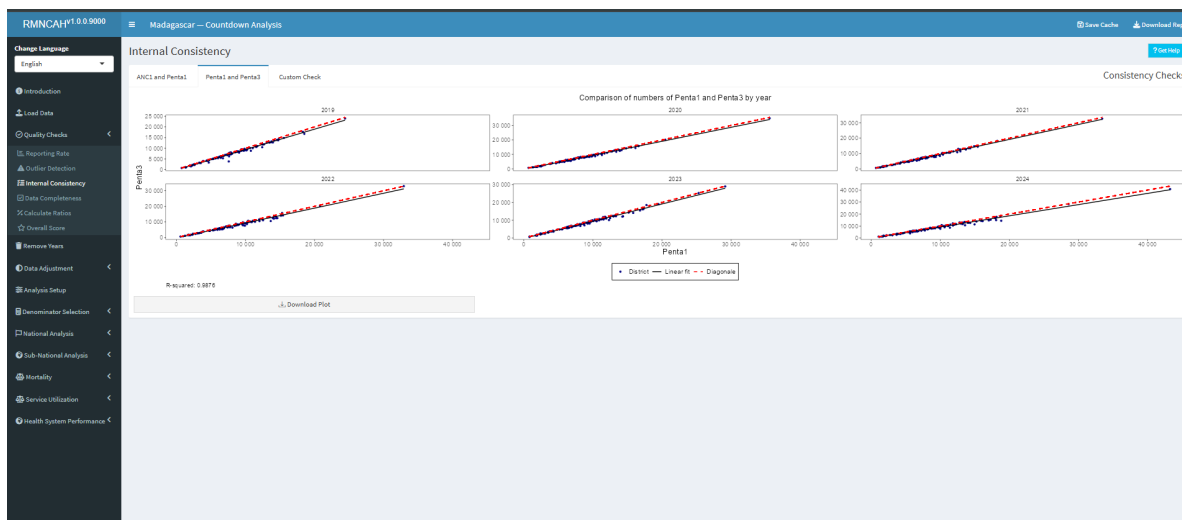
The Shiny App allows users to visually investigate which sub-national units have outliers and for which months. This way, they can factor in their country context in determining whether indeed it is a data quality issue or not.



5.2.4 Internal Consistency

Indicator	Numerator	Denominator	Consistency of Annual Interpretation
3a	N of ANC1 reported	N of penta1 reported	National ratio within expected range
3b	N of penta1 reported	N of penta3 reported	National ratio within expected range
3c	N of districts with ratios within the expected range	Total N of districts	For districts there is consistency of annual ratios
3d	N of districts within the expected range	Total N of districts	For districts there is consistency of annual ratios

There is often an inconsistency between antenatal and immunization data, even though we can argue that the two should be consistent. To examine the association between ANC1 and penta1 is particularly informative.



To compute and interpret indicators 3a and 3b the following considerations need to be made:

ANC1 to penta1 ratio

We can compute an expected ratio ANC1 to penta1 based on assumptions about mortality between early to mid pregnancy and early infancy and survey data on coverage of ANC1 and penta1 in the population:

- Consider the mortality between the first ANC visit and the first pentavalent vaccination.

Assuming that ANC1 takes place at about 20 weeks or 4-5 months of pregnancy and penta1 at 6-8 weeks postpartum, we assume a pregnancy loss (abortion) after the ANC1 visit of 3%, a stillbirth rate of 2%, a twinning rate of 1.5% and neonatal mortality rate before the penta1 of 3% then the difference between the numbers of ANC1 and penta1 should be:

$$1 - 0.03 - 0.02 + 0.015 - 0.03 = 0.935$$

This corresponds with a ANC1 to penta1 ratio of

$$1/0.935 = 1.07$$

- Actual population coverage of ANC1 and penta1 will also need to be considered, using the surveys.

The expected ratio (the number of ANC1/ number of penta1 in facilities) is $1.07 * (\text{ANC1 coverage in the survey} / \text{penta1 coverage in the survey})$. If coverage for ANC1 and penta1 are the same, then the ratio is

$$1.07(1.07 \times 1/1)$$

But if, for example, the last survey shows that ANC1 coverage was 90% and penta1 coverage was 95%, then the expected ratio becomes $1.07 * (.90 / .95) = 1.01^*$.

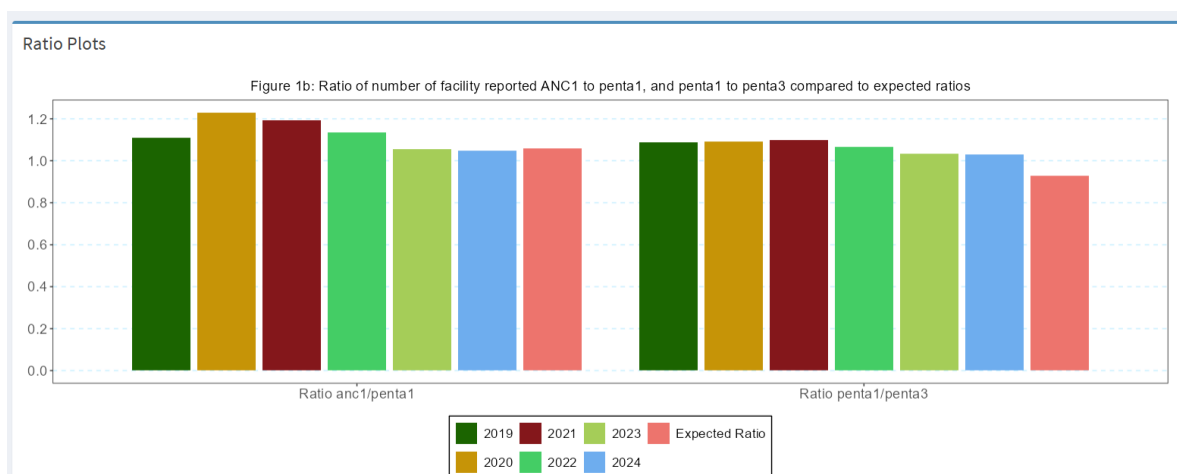
- For the national ANC1 to penta1 ratio a range of plus or minus 0.05 outside this computed ratio is considered acceptable. If the ratio is outside this range, this should be flagged, and possible explanations discussed.

Penta1 to penta3 ratio

We can compute an expected penta1 to penta3 ratio based on the most recent survey:

- The main factor determining the penta1 to penta3 ratio, which are recommended at 6 and at 14 weeks of age, is the actual drop-out rate between penta1 and penta3, as mortality plays a limited role.
- Population coverage rates from the latest survey are used to determine the expected penta1 to penta3 ratio in the facility data. For instance, if penta1 coverage is 95% and penta3 coverage is 85%, we expect that ratio to be $0.95 / 0.85 = 1.12$.
- Also, here a range of plus or minus 0.05 is considered acceptable for the assessment of the facility data.

The Figure below produced in the ShinyApp shows the ratios for all six years (2019-2024) and the expected value. It will be important to reflect on large differences (e.g., more than 0.10 or 10%).



5.2.5 Overall DQA Score

In this section, the overall DQA score is calculated by considering the mean scores within some preselected data quality metrics as below.

Summary of Performance			
Indicator	Numerator	Denominator	Interpretation
Annual data quality score (mean 1a, 1b, 2a, 2b, 3c, 3d)			

The app will automatically compute the overall data quality score based on the metrics described above. It is in this section that the first section report is downloaded capturing the key DQA indicators as shown below.

no	Data Quality Metrics	2019	2020	2021	2022	2023	2024
Type 1: Completeness of monthly facility reporting (mean of ANC, delivery, immunization, OPD)							
1a	% of expected monthly facility reports (national)	95	84	95	95	93	95
1b	% of districts with completeness of facility reporting >= 90	83	77	77	84	79	88
1c	% of districts with no missing values for the 4 forms	90	80	84	81	82	84
Type 2: Extreme outliers (mean of ANC, delivery, immunization, OPD)							
2a	% of monthly values that are not extreme outliers (national)	95	95	95	95	97	94
2b	% of districts with no extreme outliers in the year	78	85	86	84	77	76
Type 3: Consistency of annual reporting							
3a	Ratio pent1/pent1	0.81	0.95	1.05	0.98	0.92	0.92
3b	Ratio pent1/pent3	1.08	1.10	1.08	1.10	1.07	1.10
3c	% district with pent1/pent3 in expected ranged	10	33	43	37	24	25
3d	% district with pent1/pent3 in expected ranged	94	95	95	92	89	87
4	Annual data quality score	76	72	83	82	77	80

5.2.6 DQA Report and interpretation

The first report - DQA - is generated using the **Generate Report** button.

The interpretation for this section should seek to answer the following questions:

- Is there a data quality pattern by year for which there is an explanation? (include the explanation)
- Are there certain regions or other sub-national units that are particularly problematic?
- Are there certain reporting forms or services (e.g., antenatal care, labour and delivery, immunization) that are problematic?
- Is there good consistency between reported numbers of ANC1 and penta1?

¹ An extreme outlier is defined as a monthly value that is 5 times the median absolute deviation (MAD) from the monthly median value for a particular year.

6 Numerator adjustments

6.1 Rationale, Approach and implementation

Rationale: Scientific basis for the analysis

The completeness of reporting may affect the analysis, especially if completeness is low or varies between years. Extreme outliers, such as an accidental extra zero in a number, can have a large impact, especially on subnational numbers. Following the assessments, several steps are necessary to obtain a clean data set for analysis. This implies adjusting for incomplete reporting and correcting for extreme outliers.

Approach: Description of analytical steps

If we do not consider reporting completeness that means we assume all non-reporting facilities provided zero services, which is not likely to be true. Adjustments depend on how much services (e.g., pregnancy care, vaccinations) were provided at non-reporting facilities compared to those that reported. The adjustment factor k - defined as the ratio of the volume of services provided by non-reporting facilities to the volume of services provided by reporting facilities - is used to adjust the reported numbers for incomplete reporting.

6.2 Numerator Adjustment

To account for incomplete reporting, the reported number of events can be adjusted using completeness and facility reporting ratio, with the following formula:

$$N_{\text{adjusted}} = N_{\text{reported}} + N_{\text{reported}} \times \left(\frac{1}{c} - 1 \right) \times k$$

Where:

- N_{adjusted} : Total number of events adjusted for incomplete reporting
- N_{reported} : Number of events reported
- (c) : Reporting completeness (e.g., proportion of facilities that reported)
- (k) : Adjustment factor to account for lower service volume in non-reporting facilities

As a default value, we use $k = 0.25$, which means the non-reporting facilities provided services but only at a volume which was a quarter of the reporting health facilities.

The factor k can be different for different services. For instance, if private facility reporting is poor but they are in the national system and they provide a considerable number of deliveries, k maybe greater than 0.25 or even as high as 1.0.

The following k -values are used depending on the reporting used to adjust the reported numbers for incomplete reporting:

- $k = 0$ - No services in non-reporting facilities (default k -value)
- $k = 0.25$ - Some services, but much lower than reporting facilities
- $k = 0.50$ - Half the rate compared to reporting facilities
- $k = 0.75$ - Nearly as much as reporting facilities
- $k = 1.0$ - Same rate of services as reporting facilities

If the facility reporting rate is below 75%, it becomes more difficult to impute district data. Therefore, no adjustments are made if reporting is lower than 75%. In that case, further analysis to determine coverage with the facility data is not considered sufficiently reliable.

Extreme outliers, as defined in the previous section, will be corrected by imputing the median monthly value of the same year. The table summarizes the adjustments.

Table 2: Summary of adjustments made to the raw health facility data in preparation of a clean data set for the endline analysis

Problem	Action	Adjustment
Low reporting rates: identifying low rates that were adjusted	If below 75% (default), data were imputed	Median monthly value for the district year was imputed for the month with low reporting
Incomplete reporting by districts, variable over time, affecting trend assessment	If reporting rates were $\geq 75\%$ and $\leq 100\%$ (default), an assumption was made about the volume of services provided by the non-reporting facilities	Adjustment factor k value was used to adjust for incomplete reporting k default value 0.25 (replace if different value used; state if used for all reporting forms or different k factors between forms)

Problem	Action	Adjustment
Extreme outliers can greatly affect coverage trend assessments	If a monthly value was greater or smaller than 5 times the median absolute deviation (MAD) from district monthly median value, an adjustment was made	Extreme monthly outliers are corrected and given the district median value for the same year
Missing values	If there is a missing value, data were imputed	District median monthly value for the year was imputed for the month with missing value

6.3 Implementation: Conducting analysis in the Shiny App

- The outputs for this analysis can be obtained through the **Data adjustment** section of the Shiny App.
- Select the k – *factor* that is deemed appropriate given your country context.

RMNCAH1 0.0.9000 | Madagascar – Countdown Analysis | Save Cache | Download Report

Change Language: English

Data Adjustment

K-Factors (Adjustment Factors)

ANC K-Factor: 0 | Delivery K-Factor: 0 | Vaccines K-Factor: 0 | OPD K-Factor: 0 | IPD K-Factor: 0

Adjust

Adjusting the data applies custom corrections, such as removing specific years and scaling selected indicators. This process modifies the current dataset to align with predefined criteria for completeness and consistency.

Adjust Data

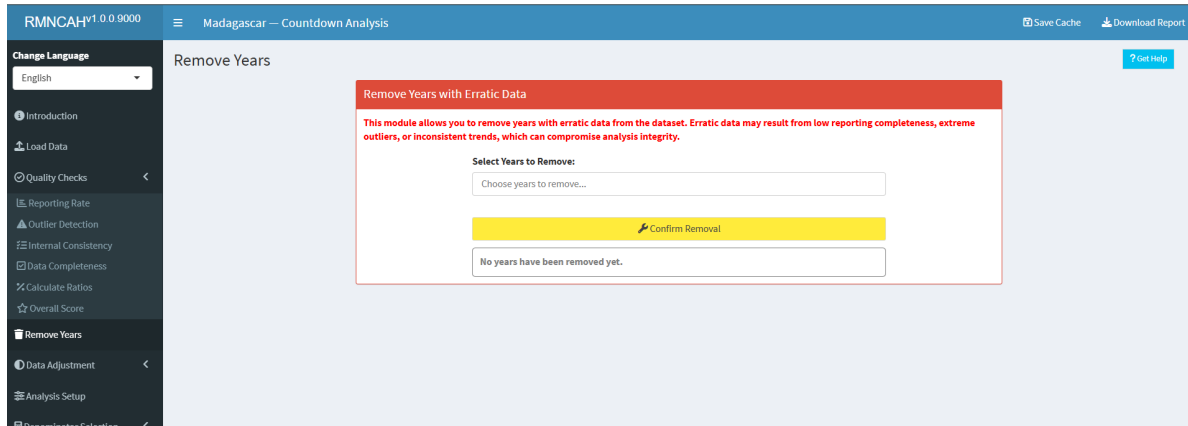
Data Adjusted

Download Adjusted Dataset

6.3.1 Remove Years

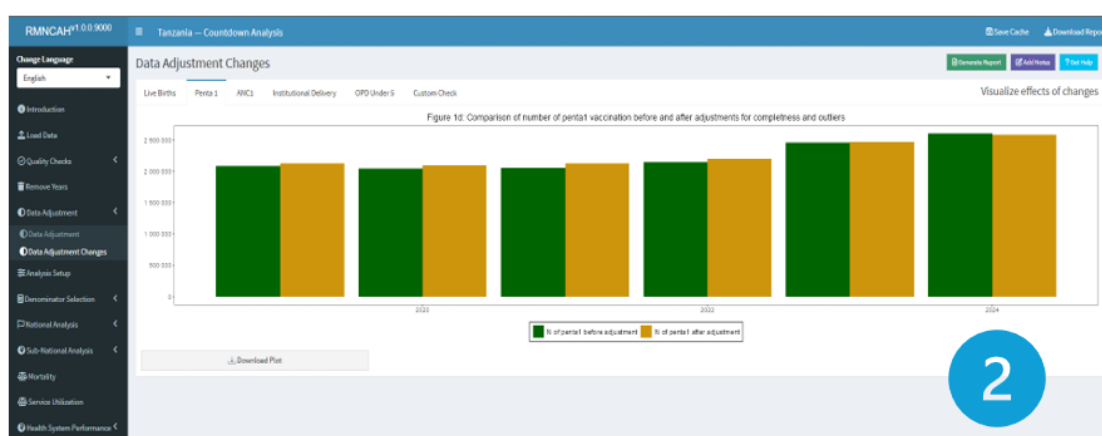
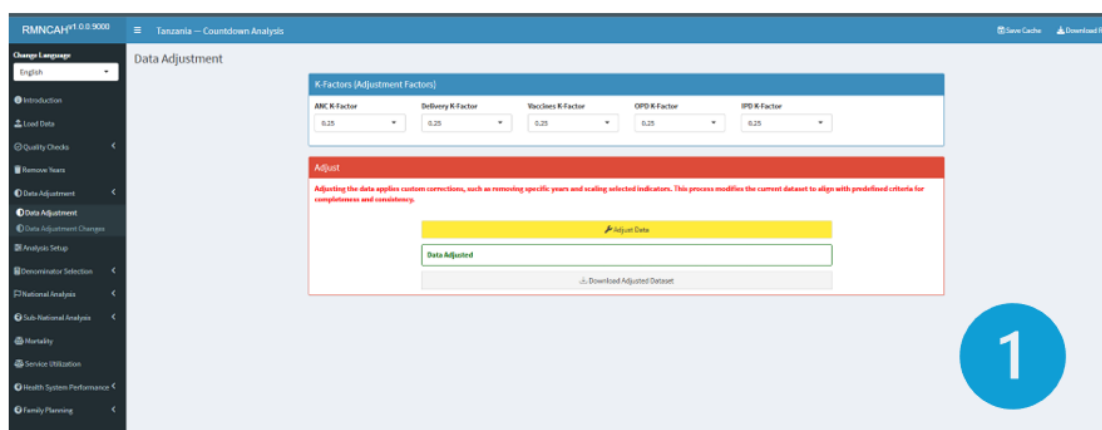
The **Remove Years** section allows you to remove any years with no data or whose data has been determined to not be fit for analysis by the country team due to quality issues.

It is recommended to be used in instances where the data quality is poor, or the country teams have sufficient, contextual information that renders the said year(s)' data unreliable/inaccurate. This will ensure that the analysis is based on reliable data.



6.3.2 Adjustment outputs

- This Section produces a master adjusted dataset (.dta) ready for analysis and plots for different indicators showing data adjustment changes. In the image below, we can see the adjustment output after making no adjustment to the data ($k = 0$).



6.3.3 Report

- The interpretation should include the selected adjustment factor (factor k) that was used to adjust for incomplete reporting (if necessary, by service). If the default factor is used, then report this and explain what this means for the reader.
- Report the percent change that the adjustment made in reported numbers of institutional deliveries and in penta1 (average of the 6-year period);
- You may want to :
 - Highlight the year with the greatest impact of the adjustment if there is one;
 - Interpret if the impact of the adjustment on coverage rates is large or small;
 - Make the same description and interpretations for penta1 vaccinations.

7 Denominator Assessment and selection

7.1 Rationale, Approach and Implementation

Rationale: Scientific basis for the analysis

Service coverage is defined as the population who received the service divided by the population who need the services (also referred to as target population). The numerators of coverage statistics (e.g., number of live births in health facilities) are derived from the health facility data and need to be adjusted as shown in the previous section. The denominator of the coverage statistics (e.g., number of live births in the population) needs to be estimated for national and sub-national levels (regions and districts).

Approach: Description of analytical steps

The objective of the health facility denominator analysis is twofold:

1. First, we assess the quality of the population projections in DHIS2 by comparison with the UN projections and the internal consistency.
2. Then, we assess the performance of multiple denominators options for the computation of population-based service coverage indicators from the health facility data.

This should lead to a final decision on denominators that are used for the analyses of population-based coverage indicators based on health facility data.

Each indicator has its own denominator, as shown in the table below.

Table 3: Selected indicators with numerators and denominators

Indicators	Numerator	Denominator
SERVICE UTILIZATION		
Outpatient visits, children under 5, per year (N)	N of OPD visits for under-5	Total mid-year population under 5
Inpatient admissions, children under 5, per year (N)	N of admissions for under-5	Total mid-year population under 5
PREVENTIVE INTERVENTIONS		

Indicators	Numerator	Denominator
% of pregnant women with 4 antenatal care visits	N of women with ANC 4th visit	Total N of pregnant women in the whole population
% of live births in health facilities	N of live births in health facilities	Total N of live births in the whole population
% of infants receiving 3 doses of pentavalent vaccine	N of infants receiving 3 doses	N of infants eligible for 3 doses of the vaccine
CURATIVE INTERVENTIONS		
% of children under 5 with malaria who receive ACT	N of children under 5 with malaria receiving ACT	Total N of children who had malaria in the last year
% of deliveries that were by C-section (population)	N of C-sections reported	Total N of deliveries in the population
% of deliveries that were by C-section (institutional)	N of C-sections reported	Total N of deliveries in health facilities
MORTALITY		
Institutional maternal mortality ratio	N of maternal deaths in health facilities	Total number of live births in health facilities
Stillbirth rate	N of stillbirths in health facilities	Total N of births in health facilities
Neonatal mortality before discharge	N of neonatal deaths before discharge (after birth)	Total N of live births in the health facilities
FAMILY PLANNING (FP)		
Ratio FP visits to women of reproductive age	N of FP new and revisits	Total N of women 15-49 years
Estimated modern use of contraceptives	Couple years of protection	Total N of women 15-49 years
FP coverage (demand satisfied)	N of women using modern methods	Total N of women in need of FP

7.1.1 Part 1: Quality assessment of population projections in DHIS2

In the first part, we assess the **quality of the DHIS2 population projections** at national level:

- **Check the internal consistency of the DHIS2 population growth over time:**

Compute the population growth rate: